2-Dimensional Solar System Simulation (inc. Launch Condition Predictions for Martian Probes), in Python

Joel Barron; s2666582

1 INTRODUCTION

Computer simulations are essential tools for modern astrophysics, used by agencies like NASA to plan missions, track near-Earth objects, and model long-term planetary evolution. This project develops a 2D, N-body simulation of the inner solar system using Python to evaluate precision of different numerical methods. We investigate three primary areas: first, assessing the impact of time-step intervals on orbital period accuracy; second, evaluating the energy conservation properties of the Beeman, Euler-Cromer, and Direct Euler algorithms; and finally, conducting a grid search to identify viable launch trajectories for Martian probes.

2 METHOD

We adopt Astronomical Units (AU), Earth Masses ($M_\oplus$), and years to make results easier to interpret, avoiding very large and very small values. SI conversion are: $1M_\oplus = 5.97217 \times 10^{24}$kg; $1\text{AU} = 1.496 \times 10^{11}$m; $1\text{yr} = 31557600$s.

2.1 Description of Simulation

Our simulation looks specifically at the Sun and the inner five planets of the solar system. The physical parameters for each body (most importantly mass) are from (Jet Propulsion Laboratory, 2019). We initialise the Sun to be at the origin and each planet on the positive x-axis, at a distance equal to their respective average orbital radius (as per (Jet Propulsion Laboratory, 2026)). We initialise velocities in the positive y-direction via the Kepler approximation. We implement the force law and Beeman algorithm, noting that we must convert G to $\approx 1.186 \times 10^{-4}\text{AU}^3\text{yr}^{-2}M_\oplus^{-1}$. The Beeman Algorithm requires acceleration at the previous time-step, which we initialise to be equal to initial acceleration. This induces a small error in calculation, but allows for simple implementation. The choice time-step, Δt, also introduces error. We will investigate this by comparing orbital periods under various choices of Δt in Experiment 1, running the simulation over a 50 year period to ensure that all planets complete multiple orbits. In general we use $\Delta t = 0.001$. We consider alternative algorithms, namely Euler-Cromer and Direct Euler, and compare their energy conservation properties. This is done in Experiment 2, over a period of 200 years to ensure long-term trends appear. Reading are taken every 20 iterations (5 per year) and analysis is done on data from one year onwards, as our initial conditions will cause artificially high potential energy.

We extend the simulation to be able to predict launch conditions and paths for Martian probes (Experiment 3). We assume that the probe is of negligible mass (does not induce gravity but is still affected by gravity of other bodies), launched at the beginning of the simulation, and is launching from just off Earth (≈ 0.001AU; this is equivalent to the probe having escaped Earth and is entering free-flight). Negligible mass allows us to simulate many satellites at once and this allows us to simulate many trajectories simultaneously to find the required launch velocity. We restrict initial x-velocity to $[0,1]$ AU/yr, and y-velocity to $[6.3, 8.3]$ AU/yr. This range keeps the magnitude of the velocities reasonable for a probe ($\lessapprox$ 10km/s relative to Earth), and ensures it will not escape the solar system. This also ensures a prograde orbit in the direction towards Mars. Assuming the probe is to perform a fly-by of Mars, we examine the simulated paths, and search for velocities which result in the minimum distance between the probe and Mars to be less than 0.006AU ($\approx 1 \times 10^6$km).

2.2 Implementation in Python

We implement the simulation using two classes. **Body** tracks kinematic state and position history via NumPy arrays.

SolarSystem manages the full simulation. We load parameters and body information from a JSON file. Orbital periods are detected heliocentrically: at each step we check whether a body has crossed from negative to positive y-position relative to the star, which itself drifts slowly over time. A softening floor ($r = 10^{-16}$ AU) prevents division by zero when calculating acceleration. We periodically write total system energy to a CSV file for analysis.

Alternative integrators are implemented as subclasses that override only *update_vectors*, adopting the respective alternative.

For simulating Martian probes, **SolarSystemSatelliteGrid** subclasses **SolarSystem** to simulate thousands of probe trajectories simultaneously. We represent probes as vectorised NumPy arrays rather than **Body** instances, making the grid search computationally feasible. We track minimum distances (to Mars) across all trajectories in a single vectorised pass per time-step.

3 RESULTS

We present results from three experiments: orbital period accuracy (Experiment 1), energy conservation across integration methods (Experiment 2), and Martian probe trajectory search (Experiment 3). Figure 1 shows a still of simulation animations, confirming qualitative accuracy in planetary orbits.

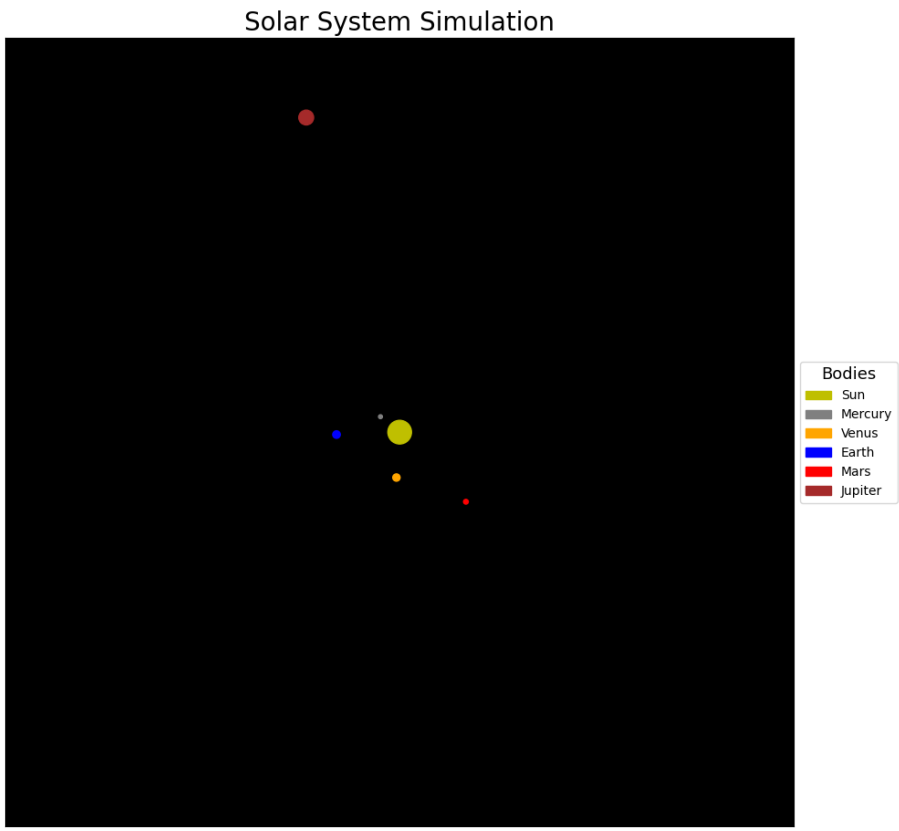

Figure 1. Still image of animation of the simulation at an arbitrary point. The Sun (yellow), Mercury (grey), Venus (orange), Earth (blue), Mars (red), and Jupiter (brown) are shown in orbit.

3.1 Experiment 1

We simulate the inner 5 planets over 50 years for our time-step choices, recording mean orbital periods. Table 1 shows the simulated values improving rapidly as Δt decreases, stabilising from $\Delta t = 0.001$ for all planets. Mercury completely fails at $\Delta t = 0.1$, but behaves well at finer time-steps, stabilising above the exact value but with the largest error of all planets ($\approx 1.12\%$). The remaining planets systematically underestimate the exact period when stabilised, with the second largest error incurred from Venus ($\approx 0.71\%$).

Planet	$\Delta t = 0.1$	$\Delta t = 0.01$	$\Delta t = 0.001$	$\Delta t = 0.0005$	Exact Value
Mercury	n/a	0.248850	0.243605	0.243566	0.240847
Venus	0.868421	0.612963	0.610938	0.610938	0.615197
Earth	1.12723	1.00122	0.999980	0.999970	1.00002
Mars	1.94000	1.87423	1.87377	1.87377	1.88085
Jupiter	11.8250	11.8345	11.8345	11.8345	11.8626

Table 1. Orbital Periods of the inner 5 planets in years for various simulation time-steps (Δt), up to 6 s.f.. Exact values from (Jet Propulsion Laboratory, 2019).

3.2 Experiment 2

We run three simulations, one for each integration algorithm. Figure 2 shows the evolution of energy over time for each algorithm. Summary statistics for each algorithm are given in Appendix A; we draw cited values from this.

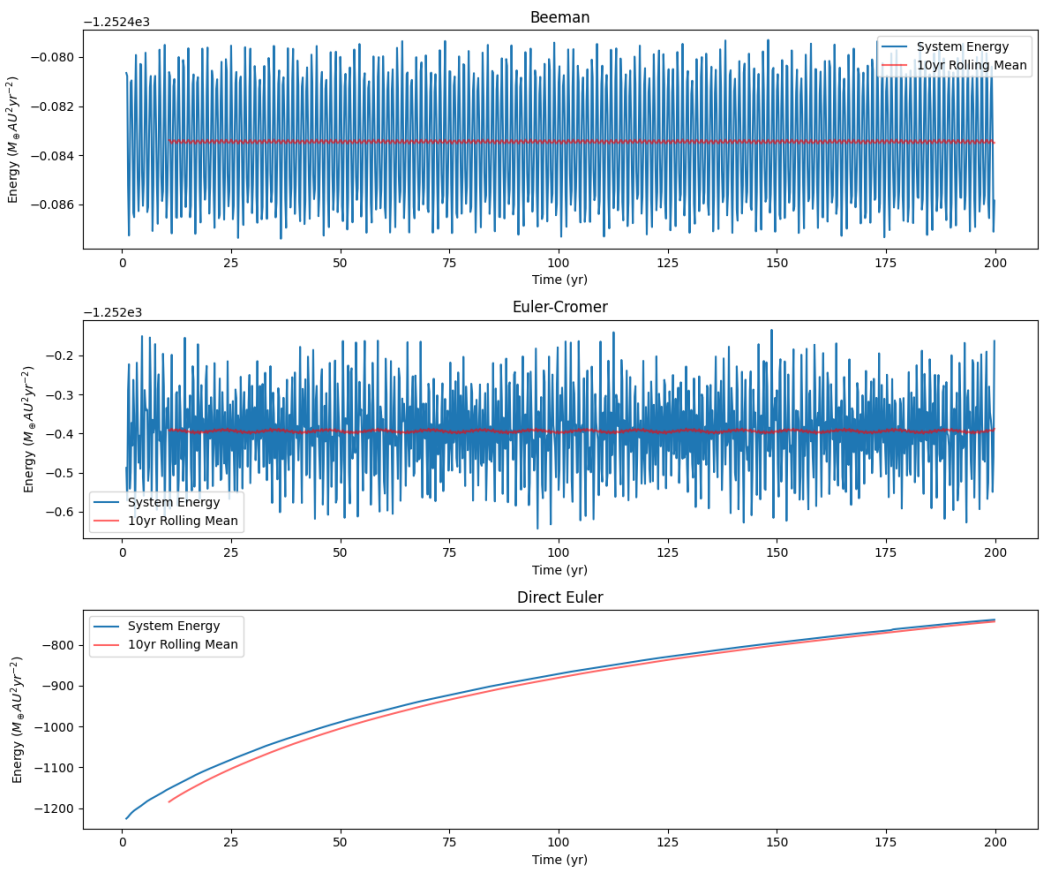

Figure 2. Subplots of system energy evolution over time with a 10 year rolling mean for different integration algorithms: Beeman (top), Euler-Cromer (middle), Direct Euler (bottom).

We observe that the Beeman algorithm produces the most stable simulation, with a maximum range of $0.0081 M_\oplus AU^2 yr^{-2}$, and a standard deviation of $0.0024 M_\oplus AU^2 yr^{-2}$. There is also essentially no linear trend over time, with a slope coefficient $\approx 1 \times 10^{-7}$. The Euler-Cromer algorithm also produces a simulation with no trend in energy over time (slope coefficient $\approx 1 \times 10^{-5}$) but is less stable than Beeman, with a range and standard deviation nearly 100 times larger. Direct Euler exhibits monotonically increasing system energy over time, with a linear slope coefficient ≈ 2.

These differences are highlighted further in Figure 3, with the normalised energy of each algorithm plotted concurrently . Direct Euler increases away from Beeman and Euler-Cromer who oscillate close to zero, with Beeman oscillating with smaller amplitude and closer to zero.

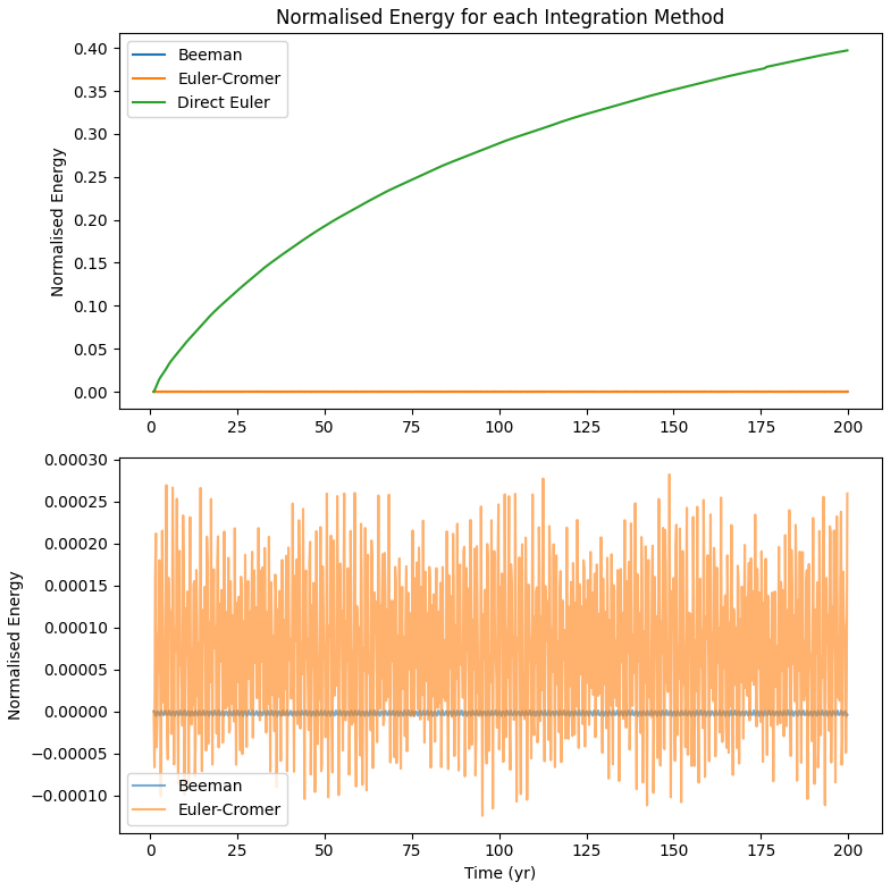

Figure 3. Subplots showing normalised energy against time for different integration algorithms.

3.3 Experiment 3

We simulate 10,000 (100×100 grid) probes with unique velocities in a reasonable range over 2 years for the outgoing journey, and 10 years to track return. We choose this as certain probes may achieve a close Mars approach on a subsequent orbit rather than their first pass. Figure 4 shows what initial conditions allow closest approach to Mars. We observe a belt of velocities around $v_y \in [7.00, 7.25]\text{AUyr}^{-1}$ across v_x that exhibits closer approach than any other initial condition.

Table 2 shows the initial velocities that allow approach of Mars within 0.006 AU ($\approx 1 \times 10^6$ km), the closest of which gets to 1.843×10^{-3}AU ($\approx 275,000$ km). All close approaches take between 1 and 1.73 years, approximately double the cruise phase of NASA's Perseverance mission (0.556 years; 203 days) (Jet Propulsion Laboratory, 2021). However, within 10 years, return distances (closest approach to Earth) are large, ranging from 0.0566AU ($\approx 8.46 \times 10^6$ km) to 0.6062AU ($\approx 9.07 \times 10^7$ km).

4 DISCUSSION

We discuss the physical interpretation of our results, sources of error, and suggested improvements.

4.1 Experiment 1

We observe convergence as $\Delta t \to 0$. This is consistent with the expected behaviour of the Beeman algorithm, a numerical approximation tending towards the exact solution as time-step decreases. Note that we use $\Delta t = 0.001$ in general for the other simulations, as convergence is well observed by this point, and computational time increases for finer Δt.

For $\Delta t = 0.1$, we observe that the orbital period of Mercury does not register. Mercury's orbital period is approximately 0.24 years, meaning that there are only 2-3 steps per orbit. As orbits are detected when the planet's position crosses from negative to positive y, this movement of Mercury is too coarse to be

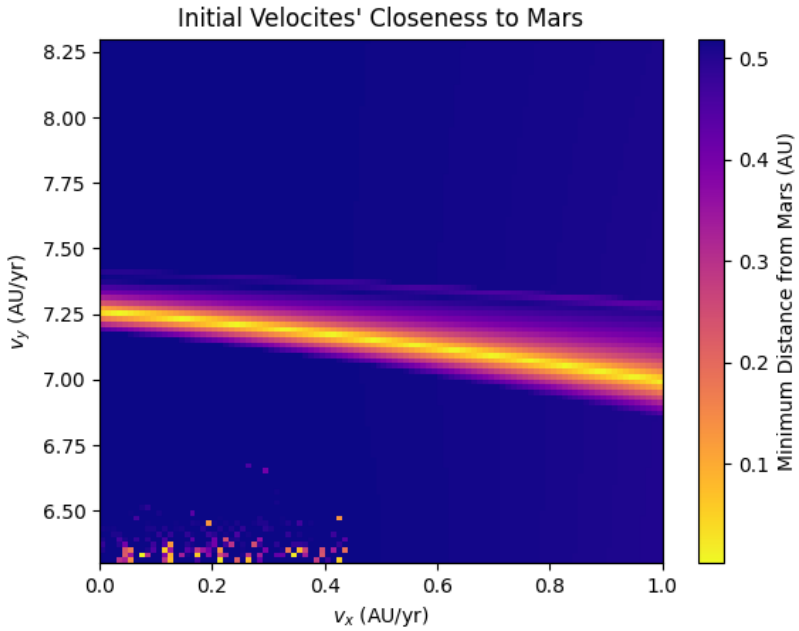

Figure 4. Heatmap showing minimum distance from Mars for each (v_x, v_y) pair.

(v_x, v_y) (AU/yr)	Dist. Mars (AU)	Time Mars (yr)	Dist. Earth (AU)	Time Earth (yr)
(0.141, 7.229)	0.001843	1.233	0.2557	4.859
(0.030, 7.249)	0.001889	1.257	0.0566	4.919
(0.768, 7.068)	0.002061	1.071	0.3684	7.764
(0.485, 7.148)	0.002683	1.149	0.3509	8.124
(0.071, 6.320)	0.003198	1.729	0.4826	9.811
(0.828, 7.047)	0.003510	1.053	0.6062	2.546
(0.242, 7.209)	0.003862	1.210	0.2618	4.851
(0.697, 7.088)	0.004359	1.091	0.3769	7.771
(0.949, 7.007)	0.005958	1.017	0.5352	2.565

Table 2. Table of the 9 probes which get within 0.006 AU ($\approx 1 \times 10^6$ km) of Mars, showing required initial velocity, closest approach to Mars, time of closest approach, the point of closest return to Earth, and the time which closest return occurs (within 10 years).

detected. Thus, this missing result is both a result of coarse time-step, and a limitation of the detection method.

We also observe systematic error in convergence. This may be due to compounding errors from the detection method. Orbital period will not be measured exactly as at each time-step as each planet discretely advances past the positive x-axis. This compounds over orbits, providing a possible explanation for larger errors seen in the inner planets, which complete more orbits in the 50 year period. Error may also be a result of our assumption of circular orbits when initialising planetary velocities when orbits are elliptical, and Mercury has the highest eccentricity ($e \approx 0.206$ (Alexander J. William Jr, 2021)) which could account for it having the largest error. However, Venus has the second largest error but the lowest eccentricity ($e \approx 0.006$ (Alexander J. William Jr, 2021)). This suggests orbital frequency and detection method may be a stronger predictor of error than eccentricity, due to the anomaly of Venus.

4.2 Experiment 2

Direct Euler is notably worse than the Beeman and Euler-Cromer Algorithms. Its monotonic increase is unphysical, as the energy should be conserved in a closed system. Direct Euler updates position using current velocity, overestimating it through each step.

The Beeman and Euler-Cromer algorithms both conserve energy over the simulation, making them both physically plausible. However, we observe that the Beeman algorithm is more stable, as it incorporates previous acceleration into position updates, reducing per-step error..

Overall, we see the importance of algorithm choice. Direct Euler creates an unphysical system, and the stability of Beeman compared to Euler-Cromer makes Beeman the clear practical choice.

4.3 Experiment 3

In section 3.3 we see a belt of velocities that allow a probe to reach Mars. Velocities in this region have appropriate energy to reach Mars's orbital radius at roughly the right time. Slower than this and the probe will fall short; faster and it will overshoot. The relative narrowness of the belt also suggests the conditions are very sensitive to the launch window. However, we also observe some velocities outside this range coming close to Mars, such as $(v_x, v_y) = (0.071, 6.320)$ seen in Table 2. This probe undershoots Mars on the first pass, but completes an orbit and approaches Mars on its second orbit, still within the 2 year simulation period.

We also note the slower journey time of our probes compared to NASA's perseverance mission. NASA operates Mars mission in a launch window that occurs roughly once every 26 months (Jet Propulsion Laboratory, 2021). Real life transfers also use techniques such as a Hohmann Transfer, which is highly efficient but dependent on precise timings and trajectories (Stefan Dubois, 2024). The precision of timing and trajectories required for Earth-Mars transfers is a major contributor to the difficult of having probes return to Earth in a reasonable time after getting close to Mars.

While we are able to find feasible trajectories and the sensitivity of launch window, the simulation is limited by the 2D plane assumption and no fuel constraints, amongst other factors.

5 CONCLUSION

This project successfully demonstrates the implementation of a 2D N-body simulation to model the inner solar system. The results indicate that the Beeman algorithm is highly effective for long-term stability, achieving minimal error in orbital period predictions and outperforming the Euler-Cromer and Direct Euler methods in energy conservation. Furthermore, the simulation identified a viable "velocity belt" for Martian transfers ($v_y \in [7.00, 7.25]\text{AUyr}^{-1}$), providing a foundational understanding of launch requirements. However, the model's precision is limited by its 2D geometry, circular orbit initialisation, and the coarse resolution of the probe velocity grid. Future iterations of this model could be extended into 3D and optimised for a finer search grid. Such improvements would allow for the analysis of more complex transfer factors, such as the timing and fuel efficiency required for a true Hohmann transfer.

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APPENDIX

Word count for this report: 1919

A. Experiment 2 Statistics

Quantitative summary statistics for system energy under each integration method, over 200 years excluding first year of simulation.

Algorithm	Mean $M_{\oplus}AU^2yr^{-2}$	STD $M_{\oplus}AU^2yr^{-2}$	Range $M_{\oplus}AU^2yr^{-2}$	Slope $M_{\oplus}AU^2yr^{-3}$	Monotone Increasing
Beeman	-1252.48345	0.002367	0.008102	-1.61914×10^{-7}	No
Euler-Cromer	-1252.39390	0.109840	0.509014	-9.52358×10^{-6}	No
Direct Euler	-901.5071	126.8088	486.7885	2.13720×10^{0}	Yes

Table 3. Summary statistics for system energy under different integration algorithms.